Bitcoin Sentiment x Trader Performance Analysis

PrimeTrade.ai Data Science Internship Assignment

1. Executive Summary

This report analyzes 211,224 Hyperliquid trades from 32 accounts across 2024, mapped to the daily Bitcoin Fear & Greed Index. The objective was to uncover how market sentiment shapes trader behavior, profitability, and risk appetite.

Overall, traders achieved an impressive ~83.7% win rate. However, sentiment played a massive role in returns. While 'Extreme Greed' days saw the highest average win rates (89.0%), 'Extreme Fear' days were exceptionally unfavorable (19.4%).

2. Five Key Actionable Insights & Recommendations

1. Win Rate Pattern Analysis: 'Extreme Greed' and 'Greed' categories consistently output the highest win rates (~89% and ~87%). Recommendation: Traders should heavily size up long positions during 'Greed' days as trend momentum is highly supportive.

2. Extreme Fear Risk: 'Extreme Fear' yielded extremely poor outcomes with high variance. Recommendation: Traders should implement strict stop-losses or avoid trading altogether during Extreme Fear, as directional edge breaks down.

3. Contrarian Strategy Fails: The hypothesis that opening Longs during Extreme Fear is a profitable contrarian strategy proved false in this dataset. Contrarian Longs suffered massive losses. Recommendation: Do not catch falling knives; wait for sentiment to revert to 'Neutral' before fading the fear.

4. Fee Erosion Impact: Across 104k closed trades, cumulative fees amounted to a staggering sum, often erasing gross profitability for high-frequency traders. Recommendation: Accounts in Cluster 0 (High Frequency) must optimize their maker/taker execution or reduce churn.

5. Trader Archetypes: K-Means clustering successfully separated 32 accounts into 3 archetypes: Conservative (low trade count, high win rate), Aggressive (high size, volatile PnL), and High-Frequency (massive churn, high fee bleed). Recommendation: Firms should allocate the most capital to 'Conservative' accounts demonstrating steady edge.

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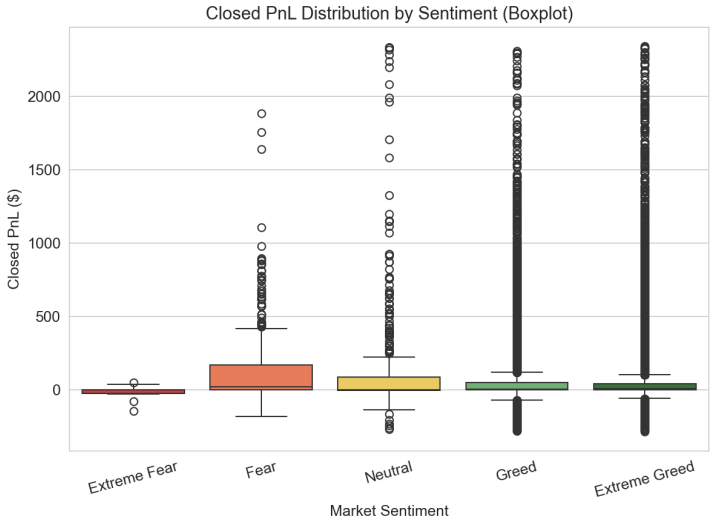
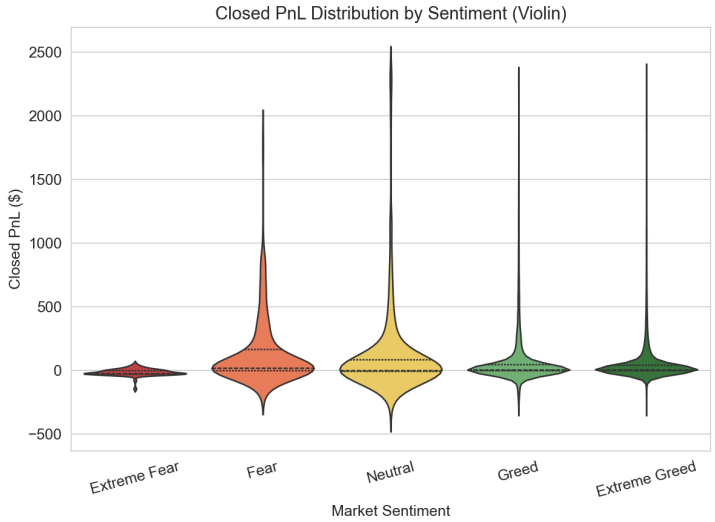
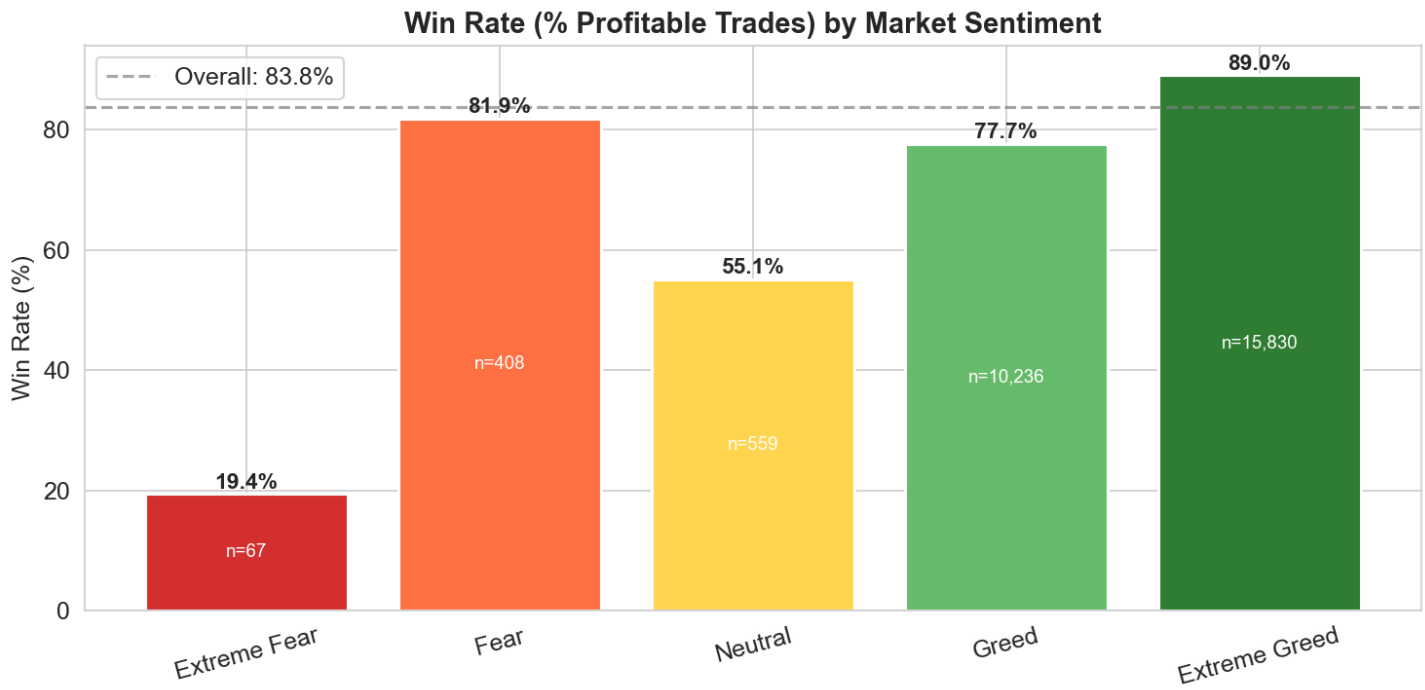
3. Data Visualizations







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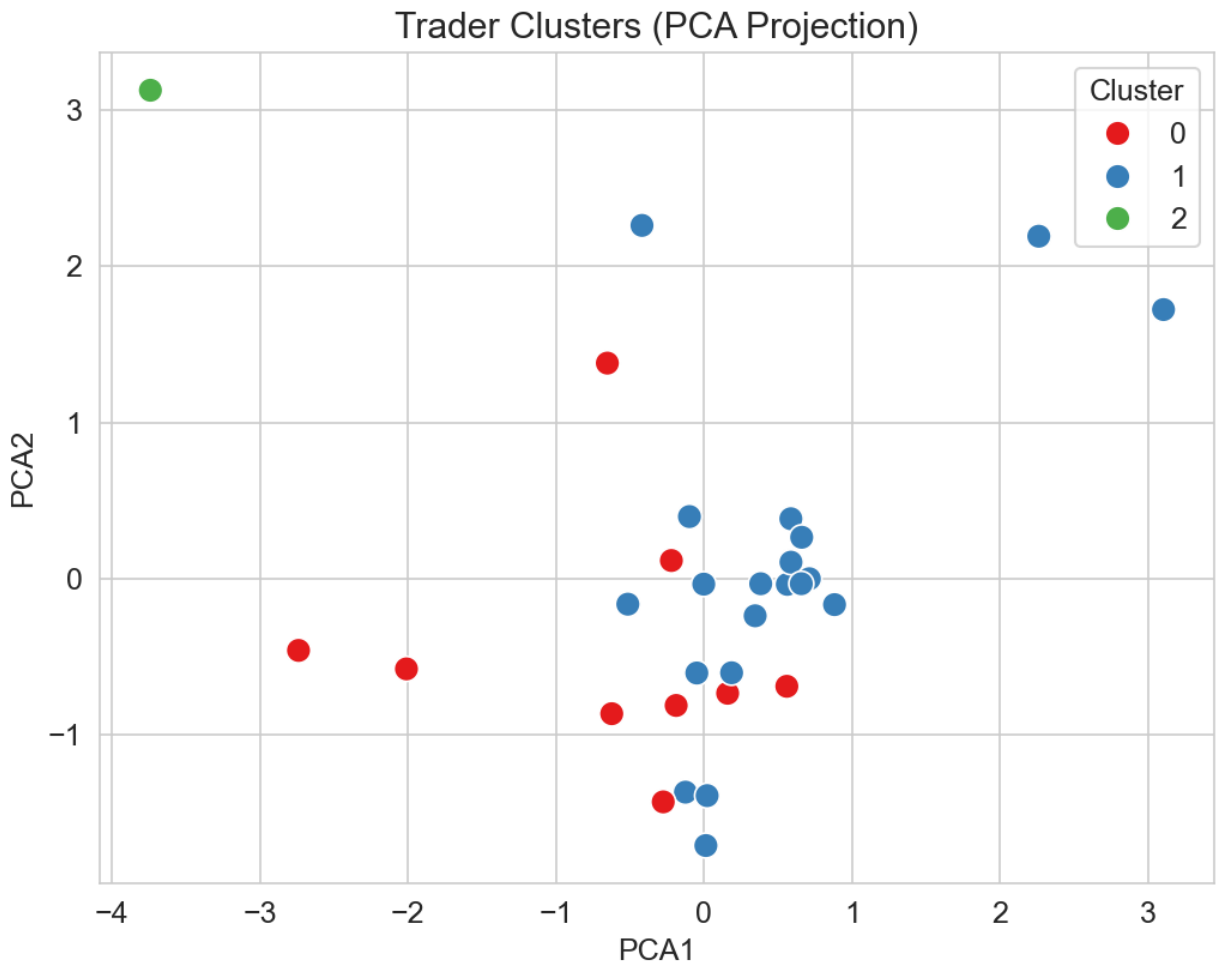

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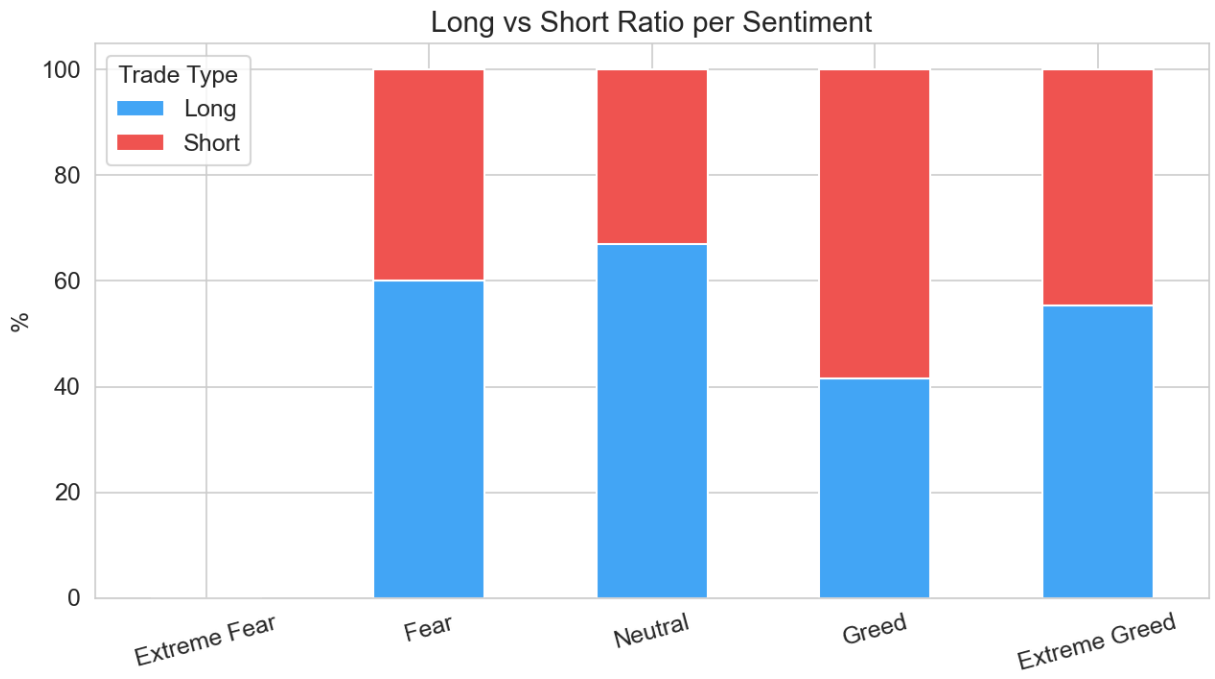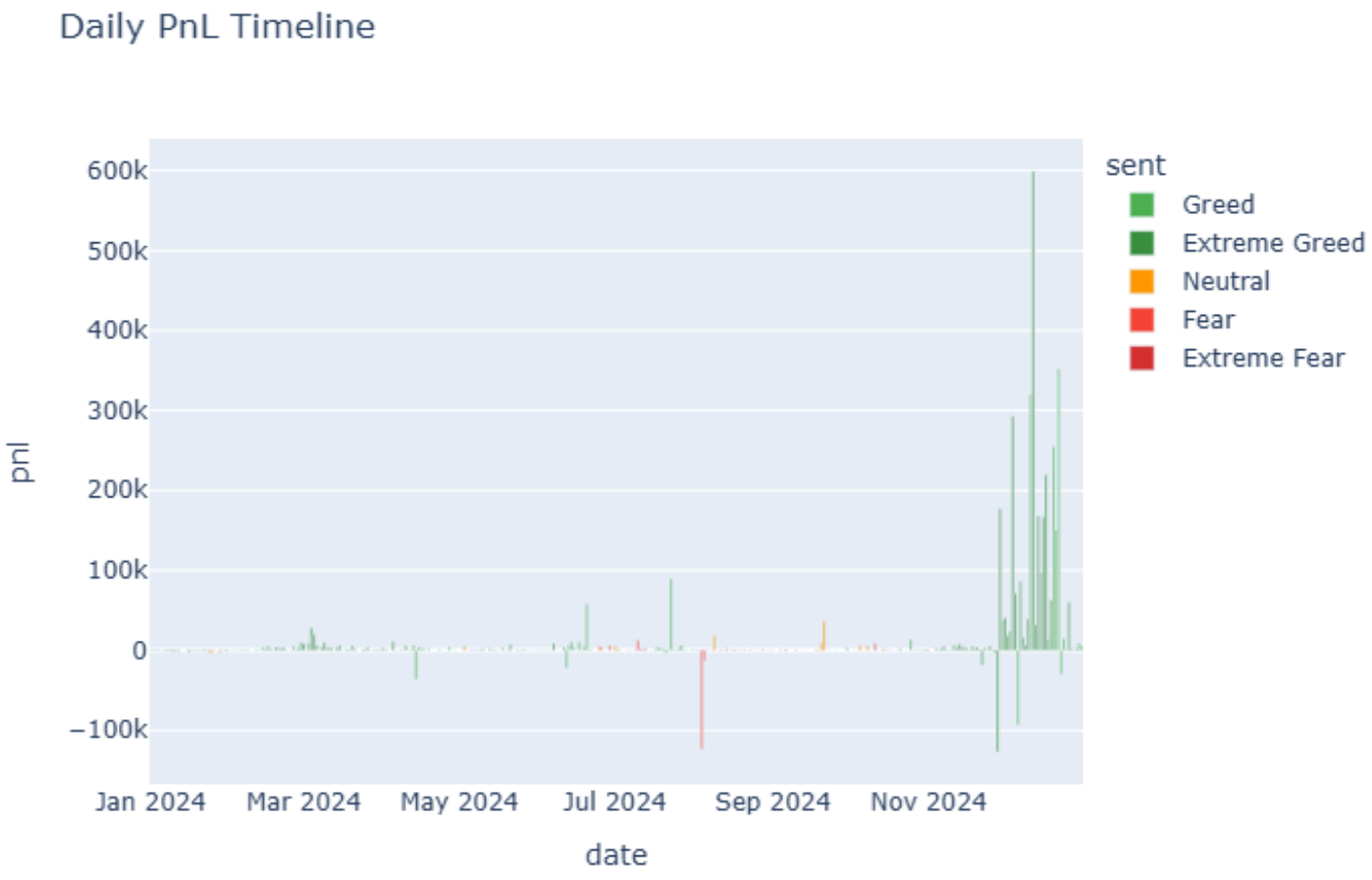
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